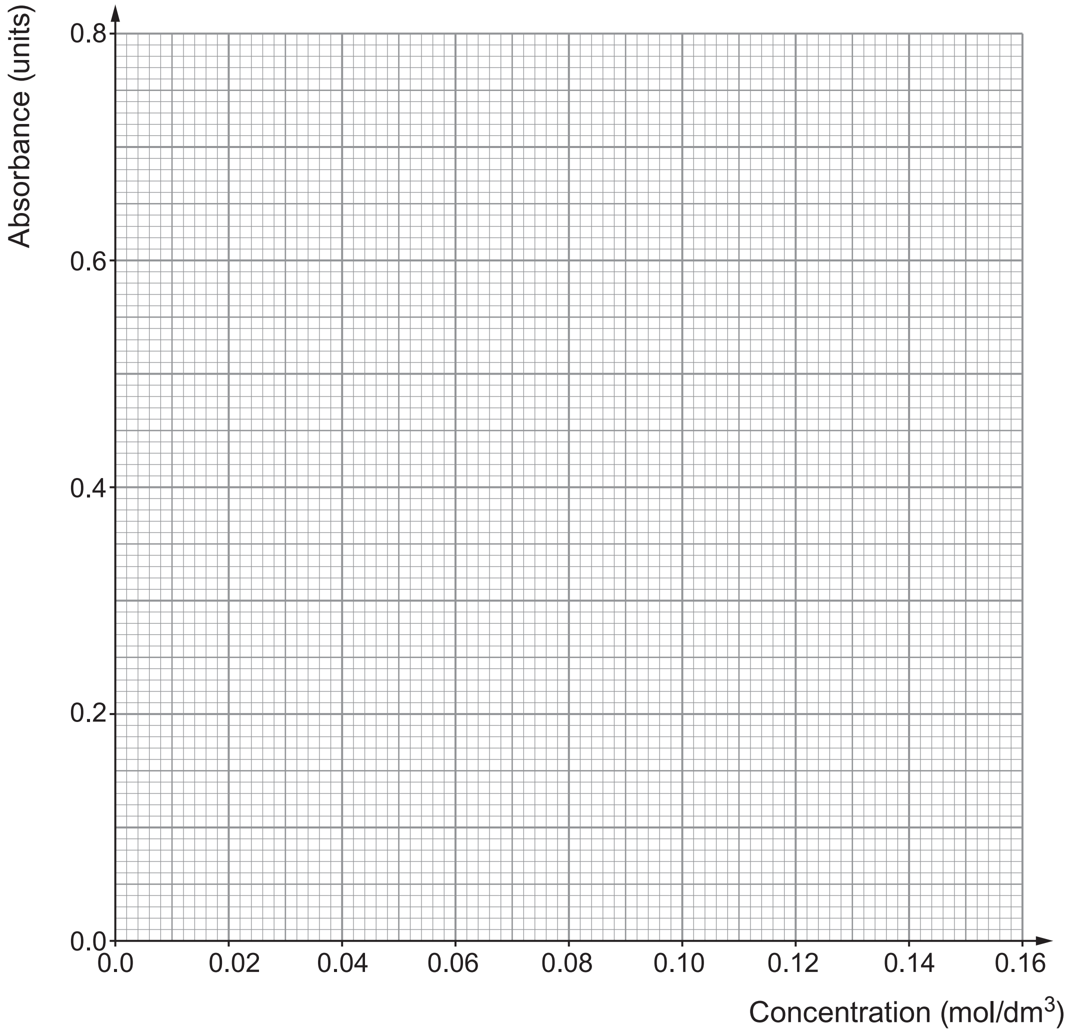
Examiner only

3. An experiment was carried out to find the concentration of beta-carotene in a solution using colorimetry. The results are shown in the table below.

Concentration (mol/dm^3)	Absorbance (units)
0.02	0.1
0.04	0.2
0.08	0.4
0.12	0.6
0.16	0.8

(a) Use the data to plot a graph on the grid below and draw a suitable line. [3]



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(b) Describe how the absorbance of beta-carotene changes with concentration. [1]

....

(c) Use your line to predict the concentration of beta-carotene with an absorbance of 0.3 units. [1]

Concentration = ... mol/dm^3

(d) It is claimed that if the concentration is doubled then the absorbance always doubles. Explain whether you agree with this statement using data from the table. [2]

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